

CUSTOMER SEGMENTATION PROJECT

Python code file

Import Libraries

```
import pandas as pd import matplotlib.pyplot as plt import seaborn as sns  
from sklearn.cluster import KMeans from sklearn.preprocessing import StandardScaler
```

Load Dataset

```
df = pd.read_csv("Mall_Customers.csv")  
print("First 5 Rows") print(df.head())  
print("\nDataset Shape:", df.shape)  
print("\nMissing Values") print(df.isnull().sum())
```

Select Features

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

Normalize Data

```
scaler = StandardScaler() X_scaled = scaler.fit_transform(X)
```

Elbow Method

```
wcss = []  
for i in range(1,11): model = KMeans( n_clusters=i, random_state=42, n_init=10 )
```

```
model.fit(X_scaled)

wcss.append(model.inertia_)

plt.figure(figsize=(8,5))

plt.plot(range(1,11),wcss,marker='o')

plt.title("Elbow Method") plt.xlabel("Number of Clusters") plt.ylabel("WCSS")

plt.show()
```

K-Means Clustering

```
kmeans = KMeans( n_clusters=5, random_state=42, n_init=10 )

df['Cluster'] = kmeans.fit_predict(X_scaled)
```

Cluster Summary

```
print("\nCluster Summary")

print( df.groupby('Cluster')[ ['Annual Income (k$)', 'Spending Score (1-100)', 'Age'] ].mean() )
```

Scatter Plot

```
plt.figure(figsize=(10,7))

sns.scatterplot( data=df, x='Annual Income (k$)', y='Spending Score (1-100)',
hue='Cluster', palette='Set1', s=120 )

plt.title("Customer Segmentation")

plt.show()
```

Count Customers in Each Cluster

```
print("\nCustomers in each Cluster")
```

```
print(df['Cluster'].value_counts())
```

Save Output

```
df.to_csv( "Customer_Segments.csv", index=False )
```

```
print("\nCustomer_Segments.csv saved successfully.")
```